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Performance of State-of-the-Art Machine Learning Methods on Out-of-Distribution Data in Tabular Regression: A Survey

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Contents

Abstract

$$P^{train} \neq P^{test}$$

1 Introduction

Artificial Intelligence (AI) [1] is a broad computer science field that studies about softwares that can mimic the human's capability of learning and problem solving. Machine Learning (ML) is a sub-field of AI, which [2] applies statistical methods to explore the underlying correlation among the features and the target. Nonetheless, ML faces difficulties when dealing with high data volume and dimensions. A sub-field of ML has been studied to overcome these challenges, i.e. Deep Learning [3], which has proven its superiority over conventional ML approaches, and yielded prospective results in Natural Language Processing (NLP), Computer Vision (CV), Large Language Models (LLMs), and also being applied to other industries ([4], [5]), e.g. healthcare, finance, energy, and agriculture.

Since 1940s [6], several ML/DL models have been proposed, and various benchmarks have been introduced to compare the performance among them. A critical fault is that these benchmarks often ([7], [8], [9], [10]) overlook the distributional shifts between the training data and deployment data, alternatively, they assume that the traing and testing sets are independent and identically distributed (i.i.d). These inherent distributional shifts are due to various factors, e.g. [8] sample biases, environment changing, data generation errors, and [11] learning spurious correlations. Many studies have been contributed to showing [11] distributional shifts in real data are roadblock to the ML/DL applications to other fields, and many [12] benchamrks and [13] AI practioners failed to analyse the models' performance on Out-of-Distribution (OOD) data. In Computer Vision, [14] showed that recent image classifiers fail to OOD generalization and [15] introduced a benchmark for performance evaluation among object detectors, namely ObjectNet, which showed performance drop of top models, from 2012 to 2018, in comparision to other standard benchmarks, e.g. ImageNet. Regarding Natural Language Processing, [16] proposed a benchmark, called CheckList, that leads to failure in semantic analysis due to tiny variations in testing texts compared to those in the training set.

Although ([17], [18]) tabular data is the primary data structure to retain data for training models in numerous areas, e.g. finance ([19], [20]), healthcare ([21], [22]) and manufacturing sectors [23], the number of standardized tabular regression benchmarks are very limited ([24], [17]). Therefore, this report serves as a benchmark consisting of various train/test splitting strategies on real datasets to evaluate model's capability towards OOD generalization on tabular regression tasks. The key contributions of our work are:

- Introduction to a comprehensive benchamrk consisting of 29 real-world low dimensional (up to 20 features) tabular datasets specifically for regression tasks. These datasets are subsequently partitioned into several train/test pairs aiming for evaluating the models' OOD generalization. To our knowledge, this marks the first benchamrk that encompasses certain approaches to split datasets for OOD extrapolation assessment.

- Conduct an experimental study to quantify and rank the robustness of a broad spectrum of standard ML/DL models on unseen distributions.

- Give insightful analysis to explore what characteristics of the proposed ML/DL methods impact their performance when generalising to unseen data regions.

Before proceeding, it is recommended to see the Appendices for the use of Abbreviations and Mathematics Notations within this report.

2 Related works

The aim of this report is to investigate the performance of widely used ML models on Out-of-Distribution (OOD) tabular data. Thus, plenty of papers were introduced to formally define what it means for models to extrapolate on unseen data. Many papers ([25], [26], [27]) have supported that extrapolation to OOD data refers to models predicting data points lying beyond the **convex hull of training samples**. Nevertheless, a more prevalent definition of OOD data highlights the **distributional shifts between the training and testing sets**, and is commonly used to compare models' generalization on OOD data ([9], [7], [12], [24], [11], [17]). To our knowledge, there are none of benchmarks for tabular regression tasks on OOD data that is built upon the first definition. Meanwhile, there are contributions, though still very limited, to benchmarking models on OOD data, based on distributional shifts, in tabular regression tasks, e.g. **Wild-Tab** [17], **Shifts Dataset** [7], and an unnamed benchmark provided by [24].

The **Wild-Tab** benchmark [17] evaluates models' robustness under covariate, subpopulation, and hybrid shifts of large-sized real-world datasets. This report also conducted an empirical study by implementing baseline DL models, e.g. MLP-PLR, FT-Transformer, and 10 other OOD generalization techniques, e.g. IRM, ERM, and GroupDRO, that are specifically designed for tabular regression tasks, and gave certain insightful analysis, i.e. ERM outperforms other models, and the impact of model architecture and hyperparameter tuning methods on models' performance. In contrast, [24] offers a benchmark to compare between the tree-based models and DL models on OOD data in tabular regression tasks. More specifically, [24] leverages 45 tabular datasets, mostly from OpenML [28], and cleans and prepares properly. Later, they benchmarked 3 tree-based models, i.e. Random Forest, XGBoost, and Gradient Boosting Trees (or HistGradient Boosting Trees in case of categorical inputs) and 4 distinct DL models, i.e. classical MLP, ResNet, FT-Transformer, and SAINT. Also, they clearly interpreted why tree-based models outperform DL models in tabular regression on OOD data regardless of models' complexity. Despite limited available benchamrks for tabular regression on OOD data, these prior works inspired us to conduct a more comprehensive empirical study to compare the performance of various ML/DL models on OOD data that are created by applying diverse splitting strategies to real-world datasets.

3 Problem settings

3.1 Regression formulation

Throughout this report, we are dealing with tabular regression tasks, let us first formally recall the formulation of this problem. To begin with, we are given an input space or domain, denoted as $\mathcal{X}$, the objective is to [29] induce an approximation function, denoted as $\hat{f} : \mathcal{X} \mapsto \mathcal{Y} \subseteq \mathbb{R}$, such that, $\hat{f} \approx f$, where f is the true function representing the relationship $\mathcal{X} \mapsto \mathcal{Y}$. To quantify how well the predictor/model $\hat{f}$ performs, several metrics have been proposed, typically Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). In practice, these evaluation metrics are leveraged as objective functions to train the models. Formally, the models repeatedly update their parameters to [11] diminish the population risk.

Definition 1. (True) Population Risk [8]

$$\mathcal{R}_{\text{true}}\left(\hat{f}\right) \equiv \mathcal{R}_{\mathbf{x}\in\mathcal{X}}\left(\hat{f}\right) \stackrel{\text{def}}{=} \mathbb{E}_{\mathbf{x}\in\mathcal{X}}\left[\ell\left(\hat{f}(\mathbf{x}), f(\mathbf{x})\right)\right]$$

Nonetheless, in real-world scenarios, it is almost impossible to capture all behaviours of $\mathcal{X}$ to train the models, e.g. self-driving car systems [11], dog/cat classification [8], and disease diagnosis for patients in all hospitals [10]. Alternatively, the model $\hat{f}$ is trained on a **sub-domain** $\mathcal{X}^{\text{tr}} \subset \mathcal{X}$, and hence, the model $\hat{f}$ is trying to optimize the sample risk or empirical risk, not the population risk as expected. In other words, while training on $\mathcal{X}^{\text{tr}} \subset \mathcal{X}$, the predictor $\hat{f}$ is expected to perform on $\mathcal{X}^{\text{te}} \subset \mathcal{X}$ as well as it does on $\mathcal{X}^{\text{tr}}$.

Definition 2. Empirical Risk [8]

$$\mathcal{R}_{\text{emp}}\left(\hat{f}\right) \equiv \mathcal{R}_{\mathbf{x}\in\mathcal{X}^{\text{tr}}}\left(\hat{f}\right) \stackrel{\text{def}}{=} \mathbb{E}_{\mathbf{x}\in\mathcal{X}^{\text{tr}}}\left[\ell\left(\hat{f}(\mathbf{x}), f(\mathbf{x})\right)\right]$$

Despite the impossibility to gather the whole domain $\mathcal{X}$, practioners and researchers have endeavoured to collect samples to form the training set $\mathcal{X}^{\text{tr}}$ that behaves somewhat similar to the whole population or domain $\mathcal{X}$. Thereby, the model $\hat{f}$ is expected to [8] make the **Empirical Risk** $\mathcal{R}_{\text{emp}}$ approximates the **True Risk** $\mathcal{R}_{\text{true}}$. Formally, this is

$$\arg\min_{\hat{f}} \mathcal{R}_{\text{emp}}\left(\hat{f}\right) \approx \arg\min_{\hat{f}} \mathcal{R}_{\text{true}}\left(\hat{f}\right)$$

To measure the similarity between the sample training set $\mathcal{X}^{\text{tr}}$ and the true population $\mathcal{X}$, we formally present two commonly used concepts, i.e. **distributional shifts** and **convex hull of data** as below.

3.2 Distributional shifts

Throughout this section, we aim to estimate the similarity between $\mathcal{X}^{\text{tr}}$ and $\mathcal{X}^{\text{te}}$ based on their statistical properties, more precisely, their **joint distributions**, i.e. $\mathbf{P}^{\text{tr}}(\mathbf{X}, \mathbf{y})$ and $\mathbf{P}^{\text{te}}(\mathbf{X}, \mathbf{y})$. More detailed, the **marginal distribution** of $\mathcal{X}^{\text{tr}}$ and $\mathcal{X}^{\text{te}}$ are written as $\mathbf{P}^{\text{tr}}(\mathbf{X})$ and $\mathbf{P}^{\text{te}}(\mathbf{X})$, respectively. On the other hand, the **conditional distribution** interprets distribution of $\mathcal{Y}$ with regards to $\mathcal{X} \mapsto \mathcal{Y}$, written as $\mathbf{P}(\mathbf{y} \mid \mathbf{X})$.

A pervasive hypothesis ([7], [8]) while training and validating models is that the train and test sets are independent and identically distributed (i.i.d). In other words, the test data $\mathcal{X}^{\text{te}}$ are considered statistically unchanged in comparision with the train data $\mathcal{X}^{\text{tr}}$.

Definition 3. No shifts - In–Distribution (ID) [8]
The test data $\mathcal{X}^{\text{te}}$ is considered to be In–Distribution (ID), or not shifted, when the following holds:

- $\mathbf{P}^{\text{te}}(\mathbf{X}) = \mathbf{P}^{\text{tr}}(\mathbf{X})$
- $\mathbf{P}^{\text{te}}(\mathbf{y} \mid \mathbf{X}) = \mathbf{P}^{\text{tr}}(\mathbf{y} \mid \mathbf{X})$

As a result, the models $\hat{f}$ show excellence performance on validation samples, but are vulnerable to real-world data. Therefore, it is imperative to detect and address these issues, so that the models can perform better on real data. In practice, there are [8] two primary types of distributional shifts, which are shifts in feature space $\mathcal{X}$, known as **covariate shifts** and shifts in label space $\mathcal{Y}$, known as **concept or semantic shift**. Both of them contribute to the fact that $\mathbf{P}^{\text{tr}}(\mathbf{X}, \mathbf{y}) \neq \mathbf{P}^{\text{te}}(\mathbf{X}, \mathbf{y})$, such test data are said to be **Out–of–Distribution (OOD)**, and thus it downgrades the performance of models $\hat{f}$. Recall, our initial objective is to induce an approximation function $\hat{f} : \mathcal{X} \mapsto \mathcal{Y}$, the mapping direction is essential to construct the upcoming definitions.

Definition 4. Covariate shifts [30]

Covariate shift is defined when the following holds:

- $\mathbf{P}^{\text{te}}(\mathbf{X}) \neq \mathbf{P}^{\text{tr}}(\mathbf{X})$
- $\mathbf{P}^{\text{te}}(\mathbf{y} \mid \mathbf{X}) = \mathbf{P}^{\text{tr}}(\mathbf{y} \mid \mathbf{X})$

Definition 5. Concept/Semantic shifts [30]

Concept shift is defined when the following holds:

- $\mathbf{P}^{\text{te}}(\mathbf{X}) = \mathbf{P}^{\text{tr}}(\mathbf{X})$
- $\mathbf{P}^{\text{te}}(\mathbf{y} \mid \mathbf{X}) \neq \mathbf{P}^{\text{tr}}(\mathbf{y} \mid \mathbf{X})$

Besides these dataset shifts, [30] theoretically there is another situation that possibly occur in real-world scenario, i.e.

- $\mathbf{P}^{\text{te}}(\mathbf{X}) \neq \mathbf{P}^{\text{tr}}(\mathbf{X})$
- $\mathbf{P}^{\text{te}}(\mathbf{y} \mid \mathbf{X}) \neq \mathbf{P}^{\text{tr}}(\mathbf{y} \mid \mathbf{X})$

3.3 Convex hull of data

Another definition, though rarely used, originated from a geometric view that leverages the definition of **Convex Hull**.

Definition 6. Convex Hull [31] [32]

Given a set of m points lying in n–dimension space, $\mathcal{S} \triangleq \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m\} \subset \mathbb{R}^n$. The Convex Hull induced by $\mathcal{S}$ is the smallest convex set that completely contains $\forall \mathbf{x} \in \mathcal{S}$. Formally:

$$\text{Conv}\{\mathcal{S}\} \overset{\text{def}}{=} \left\{ \mathbf{v} = \sum_{i=1}^{m} \lambda_i \mathbf{x}_i \mid \forall \lambda_i \geq 0 \wedge \sum_{i=1}^{m} \lambda_i = 1 \right\}$$

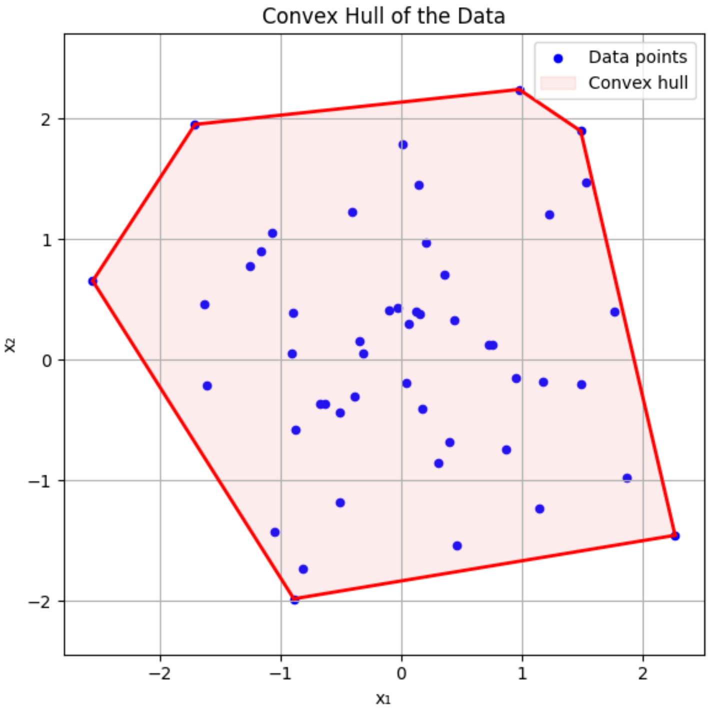

Figure 1: Convex Hull of randomly generated points in 2D visualization

Following the above definition, the OOD generalization can be interpreted as extrapolation.

Definition 7. Interpolation vs. Extrapolation with Convex Hull [25]

Given a model $\hat{f}$ trained on a set of points $\mathcal{X}^{\mathrm{tr}}$, the convex hull of $\mathcal{X}^{\mathrm{tr}}$, denoted as $\mathrm{Conv}\left(\mathcal{X}^{\mathrm{tr}}\right)$. For any point $\mathbf{x} \sim \mathcal{X}^{\mathrm{te}}$ and $\mathcal{X}^{\mathrm{te}} \cap \mathcal{X}^{\mathrm{tr}} = \emptyset$:

- Interpolation occurs when $\mathbf{x} \in \mathrm{Conv}\left(\mathcal{X}^{\mathrm{tr}}\right)$.
- Extrapolation occurs when $\mathbf{x} \notin \mathrm{Conv}\left(\mathcal{X}^{\mathrm{tr}}\right)$.

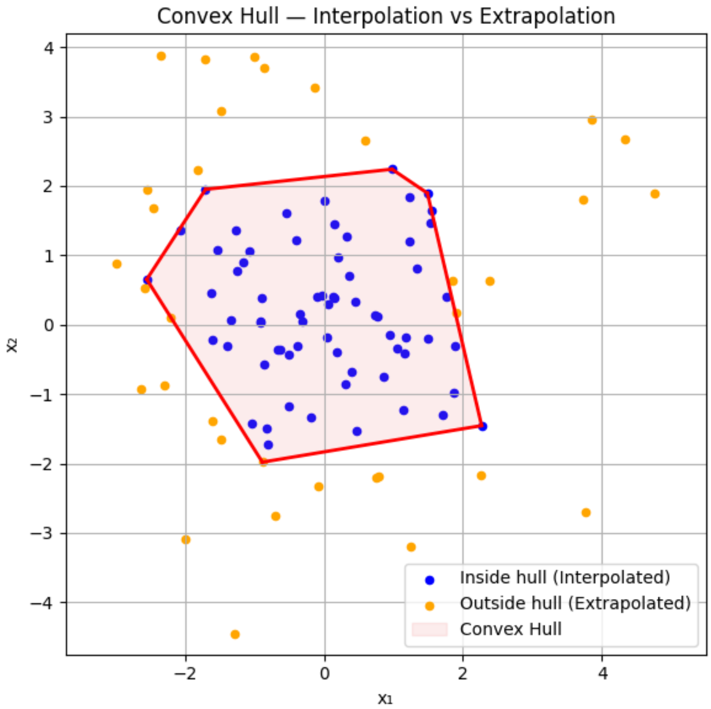

Figure 2: Interpolation vs. Extrapolation with Convex Hull in 2D visualization

In 2D space, the boundary between interpolation and extrapolation can be clearly distinguished. Nonetheless, in higher dimensional space, it is highly vulnerable to the effect of the Curse of Dimensionality [33]. More specifically, [34] the distances among the data points become more scattered, and the commonly used distance metrics, e.g. Euclidean-based, may become invalid, owing to the phenomenon of distance concentration. A (mis)conception emerged that higher precision on training set implies better models' generalization/extrapolation on unseen data regions, which was proven to be completely false, ([25], [34]), this issue is known as overfitting, which may be lessened by the expansion of number of training samples ([25], [34]). However, the datasets used within this report is low-dimensional data, at most 20 features, and the training size ranging from tens to hundreds of thoundsands samples, which may mitigate the impact of the Curse of Dimensionality.

4 Benchmark

4.1 Dataset

Original name	Dataset name	#instances	#attributes
3D Road network	3Droad	434,874	3
Airfoil self-noise	airfoil	1,503	5
Air quality	air-quality-CO	1,230	8
Appliances energy prediction	appliances-energy	19,735	27
Beijing PM2.5	beijing-pm25	41,758	11
Bias correction of numerical prediction model	temp-forecast-bias	7,752	24
Bike sharing	bike-hour	17,379	16
Blog feedback	blog-feedback	60,021	13
Buzz in social media	buzz-twitter	583,250	77
Combined cycle power plant	combined-cycle	9,568	4
Communities & crime	com-crime	1,994	122
Communities & crime unnormalized	com-crime-unnorm	2,215	126
Concrete compressive strength	compress-stren	1,030	8
Condition based maintenance of naval propulsion plants	cond-turbine	11,934	13
Cuff-less blood pressure estimation	cuff-less	61,000	2
Electrical Grid Stability Simulated	electrical-grid-stab	10,000	13

Table 1: Dataset summary

4.2 Train/Test splitting strategy

4.2.1 Random strategy

4.2.2 Distribution–based strategy

4.2.3 Geometry–based strategy

4.3 Models

5 Experimental setup

5.1 Evaluation metrics

5.2 Hyperparameters selection methods

For each traing data, the model selection method is re-run to find the most optimal hyperparameter setting w.r.t the dataset.

6 Results

7 Analysis

8 Conclusion and Discussion

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Appendices

A Notation and Abbreviation Reference List

Abbreviation

Acronyms	Meaning
tr	train
te	test
emp	empirical
vs.	versus
ID	In-Distribution
OOD	Out-of-Distribution
MSE	Mean Squared Error
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
$\mathbf{R}^2$	Coefficient of Determination

Notation

Symbol	Meaning
$\mathcal{R}(\cdot)$	Risk
$\mathcal{X}$	Input/Feature space or Domain
$\mathcal{Y}$	Output/Label space or Codomain
$\mathbf{X}^{n\times m}$	Matrix of Covariates/Features of m features and n samples
$\mathbf{y}^{n\times 1}$	Target vector of n outputs
$\mathbb{E}$	Expected Value
$\mathbf{P}(\mathbf{X},\mathbf{y})$	Joint Distribution
$\mathbf{P}(\mathbf{X}),\ \mathbf{P}(\mathbf{y})$	Marginal Distribution
$\mathbf{P}(\mathbf{y} \mid \mathbf{X})$	Conditional Distribution
$(\mathbf{x}_i, y_i) \sim \mathbf{P}(\cdot)$	A data point sampled from a distribution $\mathbf{P}(\cdot)$
$f(\mathbf{x}_i) \mapsto y_i$	True mapping from $\mathbf{x}_i \in \mathbf{X}$ onto $y_i \in \mathbf{y}$
$\hat{y}_i = \hat{f}(\cdot) \approx y_i$	An approximation mapping/function
$\ell(f(\mathbf{x}_i), y_i) \mapsto \mathbb{R}_{\geq 0}$	loss of a pair of input and true output $(\mathbf{x}_i, y_i)$
$\mathrm{Conv}(\cdot)$	Convex hull

B Datasets

C Benchmark results